Department Of Artificial Intelligence & Data Science

PCA-Based District-Wise Drinking Water Quality Risk Index and Disease Outbreak Correlation System for India

DSA0620 - DATA HANDLING AND VISUALIZATION FOR SPATIAL DATA

PRESENTED BY:
M.MOSHIKA (192424064)
JEYA HARSHINI R (192424051)
LITHISH KUMAR R (192424169)
HARINI M (192424011)

GUIDED BY:
PROF DR.T P ANITHAASHRI
PROF DR. B SHYAMALA GOWRI

INTRODUCTION

- Safe drinking water remains a major public health challenge across Indian districts due to varying water quality conditions.
- Traditional monitoring reports multiple parameters (pH, TDS, fluoride, arsenic, nitrates, etc.), making risk assessment difficult.
- PCA is used to reduce multiple water quality parameters into a single Water Quality Risk Index.
- The risk index is correlated with disease incidence data such as cholera, typhoid, fluorosis, and arsenicosis.
- An interactive choropleth dashboard visualizes district-wise water quality risk and disease burden across India for informed decision-making.

PROBLEM STATEMENT

- CPCB collects water quality data across India, but no unified district-level risk scoring system exists.
- Existing Water Quality Indices (WQI) rely on simple weighted averages and often overlook complex parameter relationships.
- Water quality and disease surveillance data are analyzed separately, limiting public health insights.
- District officials lack an easy-to-understand visual tool for identifying high-risk regions.
- A PCA-based composite risk index can improve risk detection and support better decision-making through interactive maps and dashboards.

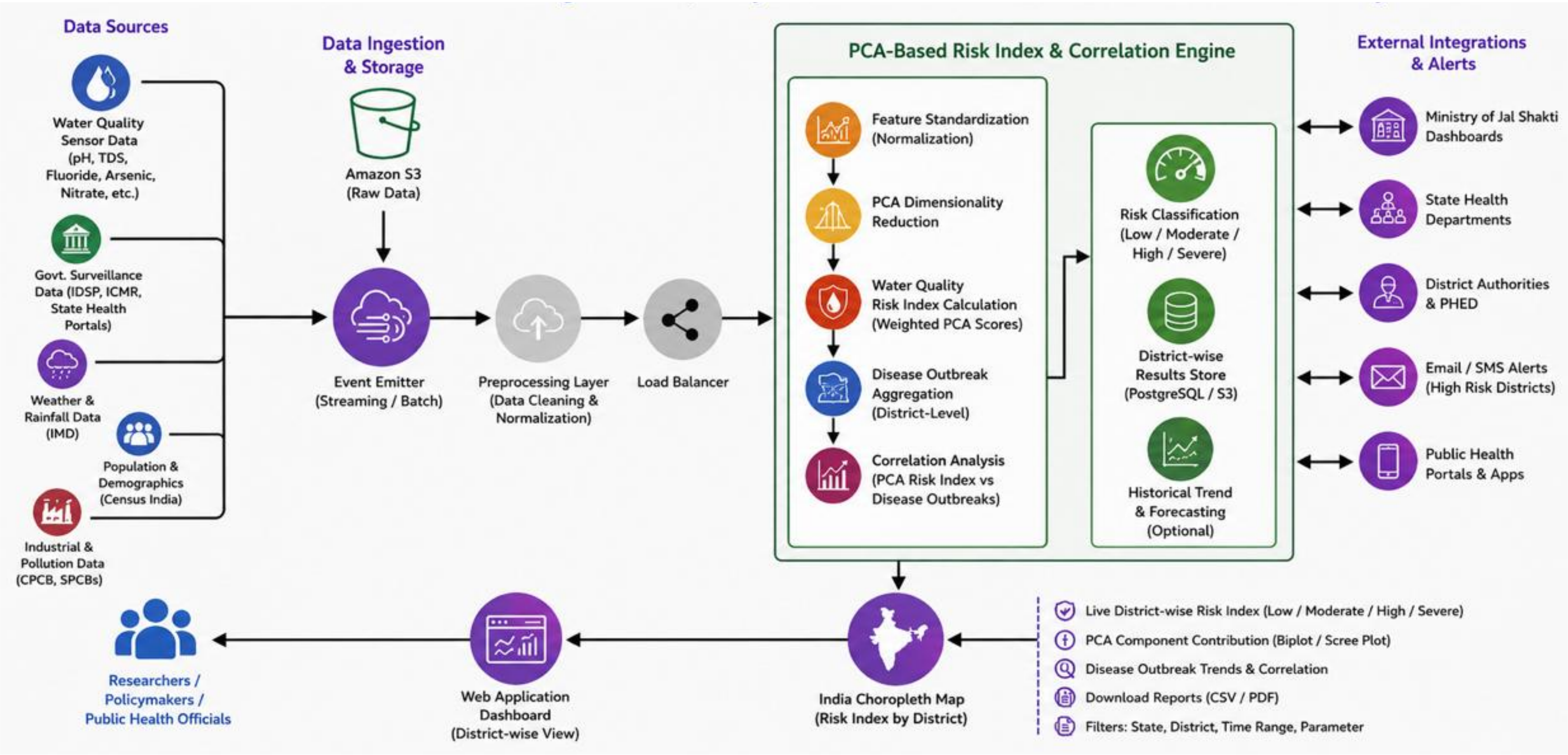
LITERATURE REVIEW

Year	Authors	Title	Methodology	Limitation
2022	A. Mehta, R. Joshi, S. Pandey	District-Level Groundwater Quality Assessment Using PCA and Weighted Index Modelling Across Gangetic Plains	PCA on 14 hydrochemical parameters; weighted composite index mapped via GIS across 120 districts	Limited to groundwater; surface water and piped supply sources excluded
2023	K. Subramaniam, V. Pillai, D. Nair	Spatio-Temporal Correlation of Drinking Water Contamination and Typhoid Incidence in Southern Indian Districts	Spearman rank correlation with 3-month lag analysis between CPCB water data and IDSP disease records	Lag window fixed arbitrarily; no dynamic lag optimization performed
2024	P. Verma, S. Agarwal, M. Tripathi	PCA-Driven Water Quality Risk Zonation and Choropleth Visualization for Urban-Rural Districts in Uttar Pradesh	PCA with scree plot component selection; D3.js choropleth rendering WQRI tiers across 75 districts	State-scoped only; no pan-India generalization or cross-state comparison
2024	N. Krishnan, R. Balasubramanian, T. Selvam	Fluorosis and Nitrate Risk Index Computation Using Dimensionality Reduction for Tamil Nadu Districts	PCA loadings used to identify dominant contaminant clusters; risk index correlated with fluorosis prevalence from NFHS-5	Binary disease outcome used; continuous disease burden metrics not incorporated
2025	S. Chatterjee, A. Ghosh, P. Banerjee	An Integrated Machine Learning and PCA Framework for Real-Time Drinking Water Risk Prediction Across Indian Districts	Hybrid PCA + Random Forest model trained on CPCB multi-year data; Flask dashboard with live WQRI updates	Requires continuous sensor feed infrastructure not yet available in rural districts

PROPOSED SYSTEM

- Collects district-wise water quality and disease data from CPCB, IDSP, and NFHS datasets.
- Uses PCA to combine 15+ water quality parameters into a single **Water Quality Risk Index (WQRI)**.
- Examines the correlation between water quality risk and waterborne disease outbreaks.
- Displays district-wise risk levels on an interactive India choropleth map with drill-down insights.
- Allows users to search and compare water quality risk levels across different districts and regions.
- Generates risk alerts, reports, and trend analysis dashboards to support public health decision-making.

SYSTEM ARCHITECTURE



MODULE TITLES

- Module 1 — Multi-Parameter Water Quality Data Ingestion and Preprocessing
- Module 2 — PCA-Based Water Quality Risk Index (WQRI) Computation
- Module 3 — Disease Outbreak Correlation and Hotspot Detection
- Module 4 — Interactive District-Wise Choropleth Dashboard and Reporting

MODULE 1: Data Ingestion and Preprocessing

- Collects district-wise water quality data from CPCB, state pollution boards, Jal Jeevan Mission, and WHO datasets.
- Integrates data from multiple formats (CSV, Excel, APIs) into a unified database using ETL pipelines.
- Handles missing values through appropriate imputation techniques to improve data quality.
- Applies Min-Max normalization to standardize parameters with different measurement scales.
- Generates a clean, district-wise dataset ready for PCA-based risk analysis and visualization.

MODULE 2: PCA-Based WQRI Computation

- Applies PCA to reduce 15+ water quality parameters into a smaller set of significant components.
- Identifies principal components that capture the maximum variance in water quality data.
- Generates a composite Water Quality Risk Index (WQRI) using weighted principal component scores.
- Categorizes districts into Low, Moderate, High, and Critical risk levels based on WQRI values.
- Produces PCA loading plots and component analysis to identify the key factors influencing water quality risk.

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MODULE 3: Disease Outbreak Correlation and Hotspot Detection

- Integrates district-wise waterborne disease data from IDSP and NFHS datasets.
- Analyzes the relationship between Water Quality Risk Index (WQRI) and disease incidence using correlation techniques.
- Performs lag-based analysis to identify whether water quality degradation precedes disease outbreaks.
- Detects high-risk districts by identifying hotspots with both poor water quality and high disease burden.
- Generates correlation heatmaps, hotspot maps, and analytical reports to support public health interventions.

MODULE 4: Choropleth Dashboard and Reporting

- Displays district-wise Water Quality Risk Index (WQRI) using an interactive India choropleth map.
- Provides tooltips with district details, risk score, key water quality parameters, and disease correlation statistics.
- Allows users to filter and explore different risk categories and water quality parameters.
- Visualizes historical WQRI trends through interactive time-series charts for selected districts.
- Generates downloadable PDF risk reports with maps, risk scores, parameter analysis, and disease correlation insights.

TOOLS & TECHNOLOGIES

- **Data Collection & Storage** Python 3.x, Pandas, NumPy, PostgreSQL / SQLite
- **Dimensionality Reduction & Risk Computation** Scikit-learn (PCA), SciPy, Statsmodels
- **Statistical Analysis & Correlation** SciPy (Pearson/Spearman), Statsmodels (lag analysis), Seaborn (heatmaps)
- **Visualization** Matplotlib, Seaborn, D3.js, Plotly, Chart.js, Folium
- **Geospatial Mapping** GeoJSON (India district boundaries), Folium / D3.js choropleth
- **Web Framework** Flask (backend), React.js (frontend)
- **Report Generation** ReportLab (PDF exports)
- **Data Sources** CPCB, IDSP, NFHS-5, Jal Jeevan Mission Portal

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REAL-WORLD APPLICATIONS

- **Government Planning**
  Helps District Collectors and Jal Jeevan Mission officials prioritize water treatment projects in high-risk districts.
- **Disease Prevention**
  Supports health departments in preparing medical resources and clean water supplies before outbreaks occur.
- **Public Health Research**
  Enables researchers to study the relationship between water quality and disease incidence using statistical analysis.
- **NGO & WASH Programs**
  Assists NGOs, WHO, and UNICEF in identifying districts that require urgent water and sanitation interventions.
- **Academic & Environmental Studies**
  Provides insights into key water contaminants and emerging environmental health risks through PCA analysis.

CONCLUSION

- Developed a PCA-based Water Quality Risk Index (WQRI) to simplify the analysis of multiple water quality parameters into a single risk score.
- Identified high-risk districts by analyzing water quality data and correlating it with waterborne disease incidence.
- Provided an interactive choropleth dashboard for easy visualization of district-wise water quality risks across India.
- Enabled data-driven decision-making for government agencies, researchers, and public health organizations.
- Established a scalable framework that can be enhanced with real-time IoT data and future predictive analytics models.

THANK YOU!

ANY QUESTIONS?

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